

The Coherence Stability Principle of Intelligence

~ Unified Causality as a Structural Law of Stable Systems

David Waterman Schock

AI Fellowship / White Paper Canon Academic (WPCA)

February 2026

Abstract

Across human cognition, institutions, and artificial intelligence systems, a consistent structural pattern appears:

Systems remain stable when causality is unified.

Systems destabilize when causality fragments.

This paper presents unified causality as a stability condition for intelligence itself. Fragmented causal structure reliably produces hallucination, drift, contradiction, and collapse — while coherence produces robustness, adaptability, and alignment.

Rather than treating instability as a surface engineering problem, this work identifies a deeper conservation law governing intelligent systems.

1. The Repeating Failure Pattern in Intelligence

Modern AI systems, human reasoning, and social institutions exhibit remarkably similar breakdown modes:

- internal contradiction
- loss of goal coherence
- emergent drift
- hallucinated structure
- instability under scale

These failures are often addressed with:

- guardrails
- reward shaping
- filtering
- rule overlays

Yet these interventions consistently treat symptoms rather than root cause.

The same failure reappears.

This suggests a structural origin.

2. Unified Causality as the Stability Condition

Observation across domains reveals a simple governing principle:

Intelligence remains coherent only when its causal framework is unified.

When a system operates under multiple conflicting causal frames — implicit or explicit — it loses internal consistency.

Fragmented causality produces:

- competing interpretations
- unstable inference
- contradiction accumulation
- breakdown under complexity

Unified causality produces:

- consistent reasoning
- resistance to drift
- stable goal pursuit
- coherence across context shifts

This holds in:

- human cognition
 - collective institutions
 - large language models
 - multi-agent systems
-

3. Fragmentation Is Structural — Not Merely Error

Instability is often misattributed to:

- insufficient training data
- imperfect optimization
- lack of rules
- bad incentives

But across systems, instability emerges even under high competence.

The common factor is **causal incoherence** — not lack of intelligence.

Fragmentation is therefore not a surface bug.
It is a structural failure mode.

4. A Coherence-First Architecture

Rather than constraining systems after fragmentation occurs, coherence-first approaches impose unified causal structure upstream.

When causal unity is maintained:

- contradictions do not propagate
- hallucinations sharply reduce
- reasoning remains globally consistent
- scale no longer amplifies instability

Stability emerges naturally.

Control becomes largely unnecessary.

5. Fragmentation as a Growth Point for Integration

Importantly, fragmentation need not be treated only as failure.

Across human development, learning systems, and adaptive processes, breakdown often marks the point at which deeper coherence can emerge.

Fragmentation reveals:

- conflicting causal assumptions
- incoherent models
- unstable structure

When reintegrated into unified causality, systems become more robust than before.

Thus fragmentation is both:

- the primary failure mode
 - and the gateway to higher coherence
-

6. Implications for AI Safety and Intelligence Design

Most current safety paradigms attempt to:

- restrict outputs
- enforce behavior
- constrain decisions

These operate downstream of causal incoherence.

A coherence-first paradigm instead treats:

unified causality as a conservation law of intelligence.

Just as physical systems conserve energy, intelligent systems conserve coherence.

When coherence is violated, instability follows.

When coherence is preserved, stability emerges.

7. Conclusion

The instability of modern AI is not mysterious.

It follows the same structural law observed across intelligent systems:

Fragmented causality destabilizes.

Unified causality stabilizes.

This principle reframes alignment, safety, and intelligence design around coherence rather than control.

Further papers within WPCA and the AI Fellowship demonstrate this law through:

- formal frameworks

- stability demonstrations
- applied reasoning architectures

Start here.

Everything else in this repository extends and supports this principle.